

Imbalance Mitigation for TCGA-UT Across Patch and WSI-Bag Regimes

Abstract

Class imbalance in computational pathology is not a single evaluation problem: labeled image patches support ordinary image classification, whereas whole-slide image (WSI) classification is commonly formulated as weakly supervised multiple-instance learning over patch-feature bags. This study evaluates imbalance mitigation on TCGA-UT with two native-regime benchmarks rather than a single cross-regime leaderboard. The patch benchmark compares cross-entropy, class weighting, focal loss, balanced sampling, Scholz-style combined losses (CE with soft F1 or soft MCC plus balanced sampling), and ProGAN-based synthetic augmentation. The WSI-bag benchmark compares matched attention-MIL baselines with RankMix and SC-MIL. All methods use slide-disjoint splits, repeated three-seed evaluation, and shared backbones within each benchmark. On patches, plain cross-entropy gives the highest macro F1 (0.314 ± 0.004), while balanced sampling gives the highest balanced accuracy (0.333 ± 0.013) at lower macro F1. On WSI bags, RankMix is the strongest method, improving balanced accuracy over plain MIL CE from 0.836 ± 0.020 to 0.863 ± 0.008 and macro F1 from 0.840 ± 0.020 to 0.848 ± 0.017 . The results show that advanced imbalance mechanisms are not uniformly beneficial, but can help when their assumptions match the native input regime.

1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive summaries. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware summaries as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

A common source of ambiguity in pathology imbalance studies is the input regime. Some methods are naturally image-level methods: they assume labeled images, image embeddings, or directly generated image samples. Other methods are naturally WSI-level methods: they assume weak labels over bags of instances and use attention, instance ranking, or bag-level representation learning. Comparing such methods in one pooled leaderboard can confound the imbalance mechanism with changes in supervision, input size, backbone, and aggregation.

TCGA-UT is useful for separating these questions because it supports two legitimate tasks with different native inputs. The dataset provides labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This study therefore uses the available supervision directly: image-level methods are evaluated on controlled patch classification, while MIL methods are evaluated on slide-level feature bags.

The contribution is a controlled benchmark design rather than a reproduction claim for every cited method. The study uses one dataset, two native tasks, and one controlled comparison within each task. Method fidelity is defined by implementing the defining imbalance mechanism in the regime for which that mechanism was designed, while holding the backbone family, split, and evaluation code fixed within each benchmark.

2 Dataset

TCGA-UT is derived from diagnostic *whole-slide images* (WSIs) in The Cancer Genome Atlas (TCGA), a pan-cancer resource that links molecular and clinical cancer data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology image dataset by extracting hematoxylin and eosin (H&E) tumor *patches* from pathologist-selected *uniform tumor regions*. It contains 1,608,060 RGB patches of size 256×256 pixels from 8736 diagnostic slides and 32 *cancer-type labels*. The patch labels are slide- or cancer-type labels, not dense tissue masks or cell-level annotations (Komura et al., 2022).

This structure makes TCGA-UT well suited for an imbalance study because the imbalance is native to the dataset rather than imposed only after sampling. The largest class is glioblastoma multiforme (GBM; 792 slides), while the smallest class is diffuse large B-cell lymphoma (DLBC; 28 slides). The resulting slide-level imbalance ratio is 28.3. As a consequence, the dataset exposes the practical tension that motivates this study: methods must learn from many visually related cancer categories while preserving performance on classes with far fewer slide-level examples.

Table 1: Dataset summary used by the shared patch and WSI-bag benchmarks.

Property	Value
Cancer-type labels	32
Diagnostic slides	8736
TCGA-UT patches	1,608,060 patches of 256×256 pixels
Largest class	Glioblastoma multiforme (GBM), 792 slides
Smallest class	Diffuse large B-cell lymphoma (DLBC), 28 slides
Slide-level imbalance ratio	28.3
Shared split size per seed	6116 train, 1310 validation, 1310 test slides
Controlled patch input	Resolution level 0, 10 deterministic patches per slide
WSI-bag input	Full frozen Virchow2 feature bags from the same slide identities

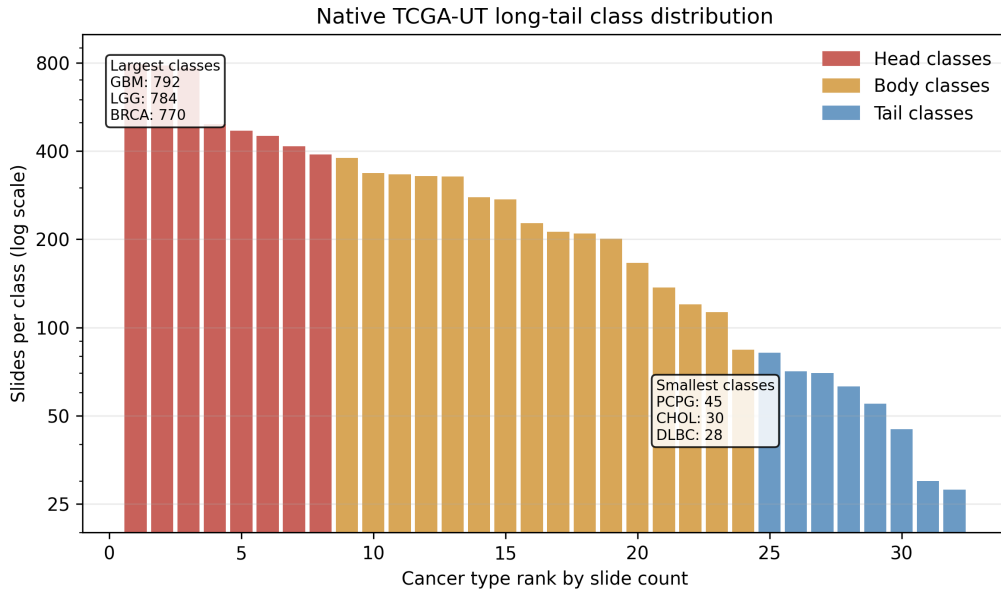


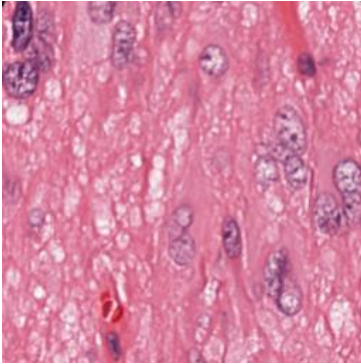
Figure 1: Native TCGA-UT slide support by cancer-type rank. The long-tailed distribution motivates metrics that weight classes more evenly than aggregate accuracy.

2.1 Data Units and Terminology

Several data units are involved in the benchmark, and the distinction matters for method comparison. A *slide* or *WSI* is one diagnostic histopathology image associated with one cancer-type label. A *tumor region* is a pathologist-selected area of a slide from which TCGA-UT crops groups of patches. A *patch* is one 256×256 RGB crop used directly by the patch classifier. A *feature chunk* is a saved frozen-feature representation associated with a slide or slide subset. A *WSI bag* is the collection of frozen patch-feature instances used by the attention-MIL model for one slide-level prediction. *Head*, *body*, and *tail* classes refer to relative class-support tiers in the long-tailed slide distribution. A *slide-disjoint split* means that all patches and features from one slide stay in exactly one of train, validation, or test.

Table 2: Terminology used for the two native input regimes.

Term	Meaning in this study
<i>Slide / WSI</i>	Diagnostic whole-slide image carrying one cancer-type label.
<i>Patch</i>	256×256 H&E crop from a selected tumor region; patch labels inherit the slide cancer type.
<i>Region</i>	Tumor area selected before patch extraction.
<i>Cancer type</i>	The 32-way class label used for patch and slide-level classification.
<i>Feature chunk</i>	Frozen Virchow2 feature representation used to assemble slide-level feature bags.
<i>WSI bag</i>	Set of patch-feature instances for one slide, consumed by the MIL model.
<i>Head/body/tail class</i>	Class-support tier derived from the slide-count distribution.
<i>Slide-disjoint split</i>	Split rule preventing any slide’s patches or features from crossing train, validation, and test.



Field	Example value
Cancer type	Glioblastoma multiforme
Resolution level	0
Patch size	256×256 RGB pixels
Source unit	One crop from a selected tumor region
Patch role	Image-level sample if selected by the controlled manifest
WSI-bag role	Candidate instance in the slide’s feature bag

Figure 2: Example TCGA-UT data point copied from the shared raw dataset. The image is a single patch, while the table shows how that patch relates to the manifest fields used by the benchmark.

2.2 Shared Splits and Benchmark Inputs

For each seed, slides are split into 6116 training, 1310 validation, and 1310 test slides. The same slide-level assignments are reused by both benchmarks so that no patches from the same slide can appear in different splits. This is the main leakage-control mechanism: the patch benchmark and the WSI-bag benchmark see different input units, but they inherit the same train/validation/test slide partition.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples 10 patches per slide with a fixed rule before method-specific training begins. This makes the patch benchmark a controlled comparison of imbalance

mechanisms rather than a comparison of patch-selection policies. The WSI-bag benchmark uses full frozen Virchow2 feature bags built from the same slide identities, rather than applying an additional instance cap (Zimmermann et al., 2024).

3 Methods

The study organizes methods into four imbalance-mitigation categories: data-level augmentation, synthetic generation, algorithm-level losses, and representation or architecture methods. The central control is comparability within regime. The patch benchmark uses one shared patch-classification backbone, and the WSI-bag benchmark uses one shared attention-MIL backbone. A method is named after a cited paper only where its defining imbalance mechanism is implemented in the corresponding native regime.

3.1 Patch-Level Benchmark

Table 3: Patch-level benchmark methods.

Category	Method	Implemented mechanism
Baselines	CE, weighted CE, focal loss, balanced sampling	Shared patch classifier with objective or sampler changes only.
Algorithm-level	CE + soft F1/MCC (balanced)	Equal-weight sum of cross-entropy with macro soft F1 or multiclass soft MCC, trained with class-balanced oversampling (Scholz et al., 2024).
Synthetic generation	ProGAN augmentation	Class-specific progressive generators balance minority patch classes to the training-set head count; this is a Ruiz-Casado-style ProGAN adaptation from binary to multiclass histology (Ruiz-Casado et al., 2024).

The patch baselines isolate common imbalance interventions. Weighted CE changes the class contribution to the loss, focal loss down-weights easy examples, and balanced sampling changes minibatch composition without changing the loss. The Scholz representatives follow the authors’ strongest configuration: class-balanced oversampling together with an equal-weight combination of cross-entropy and a differentiable macro soft-F1 or multiclass soft-MCC objective. ProGAN augmentation trains class-specific generators for minority classes and augments the patch training set; generated patches are used only as additional patch-classification training examples, not as synthetic slide bags.

3.2 WSI-Bag Benchmark

The WSI baselines use the same attention-MIL architecture and frozen feature bags. RankMix first trains a teacher MIL model, derives ranked instance subsets from the teacher attention and pseudo-label signal, and then trains with mixed representative feature subsequences and soft labels. SC-MIL adds a supervised-contrastive bag representation objective through a projection head and linearly shifts training weight toward the classifier objective over time. Activation diagnostics are recorded for both methods to verify that augmentation and contrastive pair construction occur during training.

Table 4: WSI-bag benchmark methods.

Category	Method	Implemented mechanism
Baselines	MIL CE, weighted MIL, focal MIL, balanced MIL	Shared attention-MIL backbone with objective or sampler changes only.
Data-level augmentation	RankMix	Two-stage attention-MIL training with teacher-derived patch pseudo-labels, ranked representative feature subsequences, and soft-label feature mixup (Chen and Lu, 2023).
Representation	SC-MIL	Bag-level supervised contrastive learning with a projector head and a linear curriculum from representation to classifier learning (Juyal et al., 2024).

4 Evaluation

Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, head/body/tail summaries, negative log likelihood, multiclass Brier score, and expected calibration error are also computed where probabilities are available. Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported tables use the held-out test split and show mean $\pm$ standard deviation over seeds. Separate validation summaries are retained as artifacts for model-selection checks. The completed sweep contains no missing planned results for either benchmark.

5 Results

5.1 Patch-Level Results

Table 5: Patch-level test results over three seeds. Methods share the same patch manifest, resolution, classifier family, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
CE	0.412 ± 0.011	0.310 ± 0.003	0.314 ± 0.004
Focal	0.402 ± 0.010	0.307 ± 0.011	0.310 ± 0.011
ProGAN augmentation	0.407 ± 0.012	0.306 ± 0.021	0.299 ± 0.021
Balanced sampler	0.354 ± 0.011	0.333 ± 0.013	0.293 ± 0.006
Weighted CE	0.333 ± 0.033	0.330 ± 0.016	0.285 ± 0.018

Plain CE is the strongest patch method by the primary metric, with macro F1 0.314 ± 0.004 and accuracy 0.412 ± 0.011 (Table 5). Focal loss is close, with macro F1 0.310 ± 0.011 . Balanced sampling and weighted CE shift the operating point toward rare-class sensitivity: they produce the highest balanced accuracies, 0.333 ± 0.013 and 0.330 ± 0.016 , but lower macro F1 and accuracy. This tradeoff indicates that the class-balancing interventions improved average recall but introduced enough precision or head-class loss to reduce the harmonic classwise summary.

Among the representative advanced patch methods, ProGAN augmentation reaches macro F1 0.299 ± 0.021 , below CE and focal loss (Table 5). The Scholz combined losses are reported in the same table; their macro-F1 and balanced-accuracy rankings should be read against both plain CE and balanced-sampling CE, because the Scholz configuration deliberately couples metric-derived objectives with oversampling. The result should not be interpreted as a general

rejection of synthetic generation or metric-derived losses. It is narrower: under this fixed patch set, classifier family, and three-seed configuration, not every imbalance mechanism that helps in other medical imaging settings transfers to TCGA-UT patches.

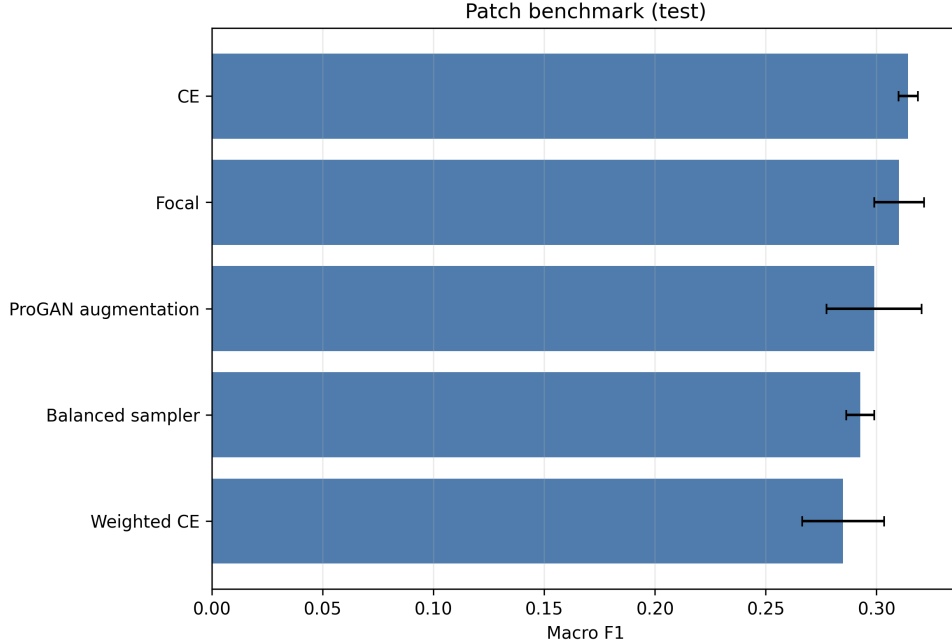


Figure 3: Patch-level macro F1 on the held-out test split. The figure visualizes the same ranking as Table 5; absolute values should only be interpreted within the patch benchmark.

5.2 WSI-Bag Results

Table 6: WSI-bag test results over three seeds. Methods share the same full feature bags, attention-MIL backbone family, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
RankMix	0.894 ± 0.010	0.863 ± 0.008	0.848 ± 0.017
MIL CE	0.890 ± 0.006	0.836 ± 0.020	0.840 ± 0.020
Balanced-sampling MIL	0.885 ± 0.007	0.837 ± 0.015	0.836 ± 0.013
SC-MIL	0.881 ± 0.007	0.837 ± 0.017	0.833 ± 0.016
Weighted MIL CE	0.880 ± 0.005	0.834 ± 0.018	0.829 ± 0.018
Focal MIL	0.879 ± 0.004	0.824 ± 0.011	0.827 ± 0.009

RankMix is the strongest WSI-bag method by all three aggregate metrics (Table 6). Relative to plain MIL CE, it improves balanced accuracy from 0.836 ± 0.020 to 0.863 ± 0.008 and macro F1 from 0.840 ± 0.020 to 0.848 ± 0.017 . The macro-F1 gain is modest, but the balanced-accuracy gain is larger and consistent with RankMix directly augmenting representative instance subsequences in the MIL regime.

The remaining WSI methods cluster closely. Balanced-sampling MIL and SC-MIL produce similar balanced accuracy (0.837), but both trail RankMix in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not improve over plain MIL CE in this configuration, with focal MIL giving the lowest WSI macro F1. Activation diagnostics confirm that the advanced WSI

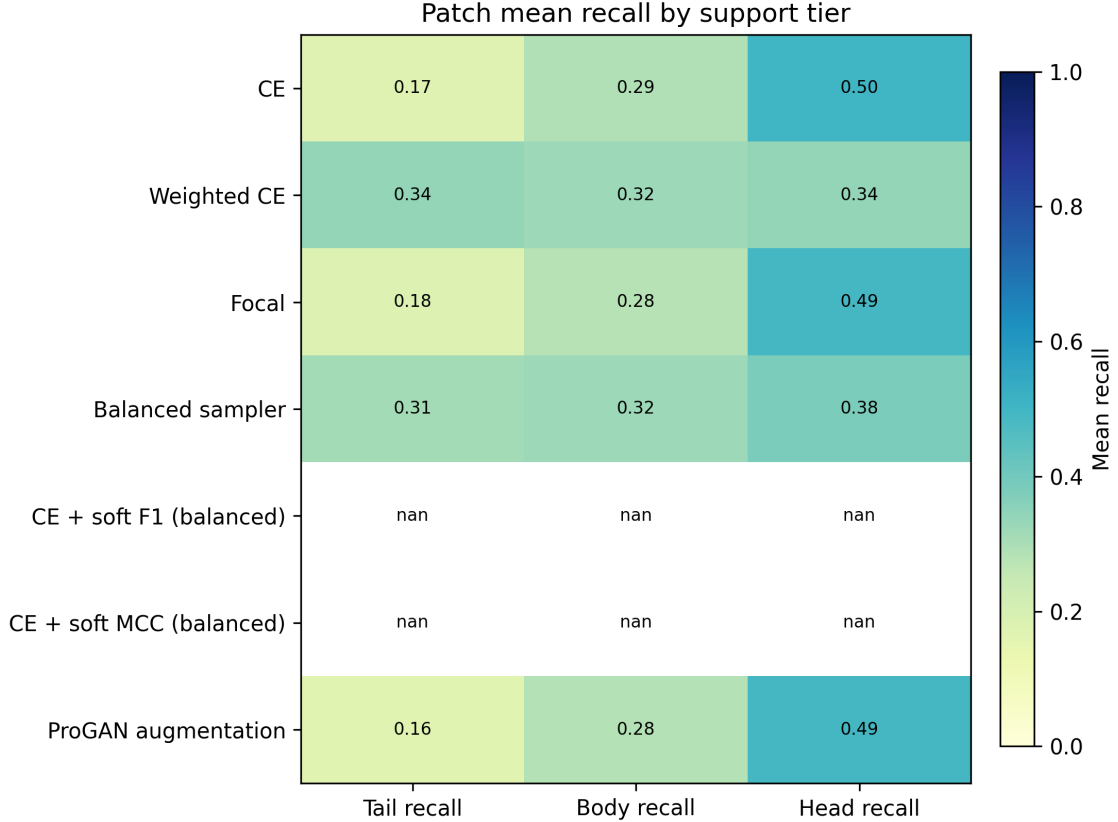


Figure 4: Patch-level classwise recall on the held-out test split. The plot exposes the class-level variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

objectives were active: RankMix generated about 1.10 million mixed examples per seed, and SC-MIL constructed about 1.06 million positive pairs per seed.

6 Discussion

The two benchmarks support different conclusions. In the patch benchmark, simple CE and focal loss are hard to beat by macro F1, while class-balanced sampling and weighting mainly tradeoff macro F1 for balanced accuracy; metric-derived Scholz losses are evaluated in that same tradeoff space because they are trained with balanced sampling. In the WSI-bag benchmark, RankMix improves the strongest baseline and is the clearest case where a more elaborate imbalance-specific mechanism helps. The same study therefore mixes negative and positive transfer results across patch-level synthetic or metric-derived methods and WSI-native feature mixing.

This pattern reinforces the need to evaluate imbalance methods in their native input regimes. A patch generator should not be judged by first converting synthetic images into one-instance MIL bags, and a MIL feature-mixing method should not be compared directly against a patch classifier. The controlled design narrows the claims but makes them easier to defend: within each benchmark, the split, backbone family, input budget, and evaluation code are shared.

Several limitations remain. First, the study is a controlled benchmark, not a full reproduction of every original paper. The ProGAN experiment keeps the progressive-generator mechanism from Ruiz-Casado-style histology augmentation but adapts it from binary breast histology to multiclass TCGA-UT and does not reproduce the full GAN-family comparison. Second, TCGA-UT patch labels come from pathologist-selected tumor regions; they are use-

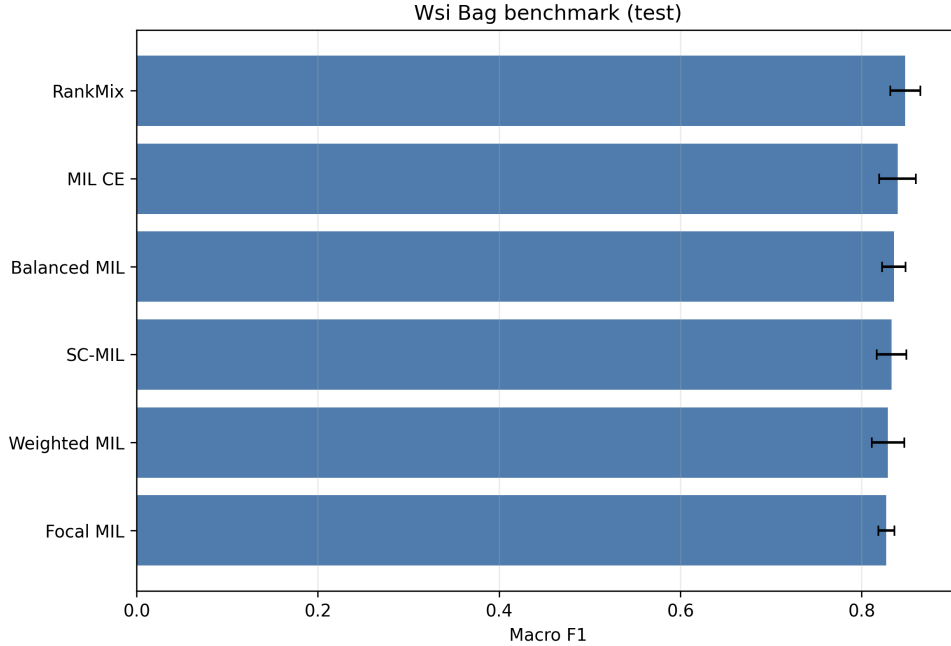


Figure 5: WSI-bag macro F1 on the held-out test split. RankMix is the top method within the WSI-bag benchmark, but the absolute scale is not comparable to the patch benchmark because the task input differs.

ful patch-level class labels but not dense annotations. Third, the WSI benchmark uses frozen Virchow2 feature bags and an attention-MIL family, so conclusions apply to this representation setting. Finally, advanced methods may benefit from additional hyperparameter tuning beyond the fixed three-seed benchmark used here.

7 Conclusion

TCGA-UT supports a cleaner imbalance study when patch-level and WSI-bag supervision are evaluated separately. On controlled patch classification, the strongest macro-F1 baseline is plain CE, and balancing interventions mainly change the recall-precision tradeoff. On WSI bags, RankMix provides the strongest overall result and the clearest improvement over a plain MIL baseline. The practical implication is that imbalance mitigation should be selected and evaluated at the same level where the model actually consumes data; otherwise, apparent method differences can be artifacts of incompatible task regimes.

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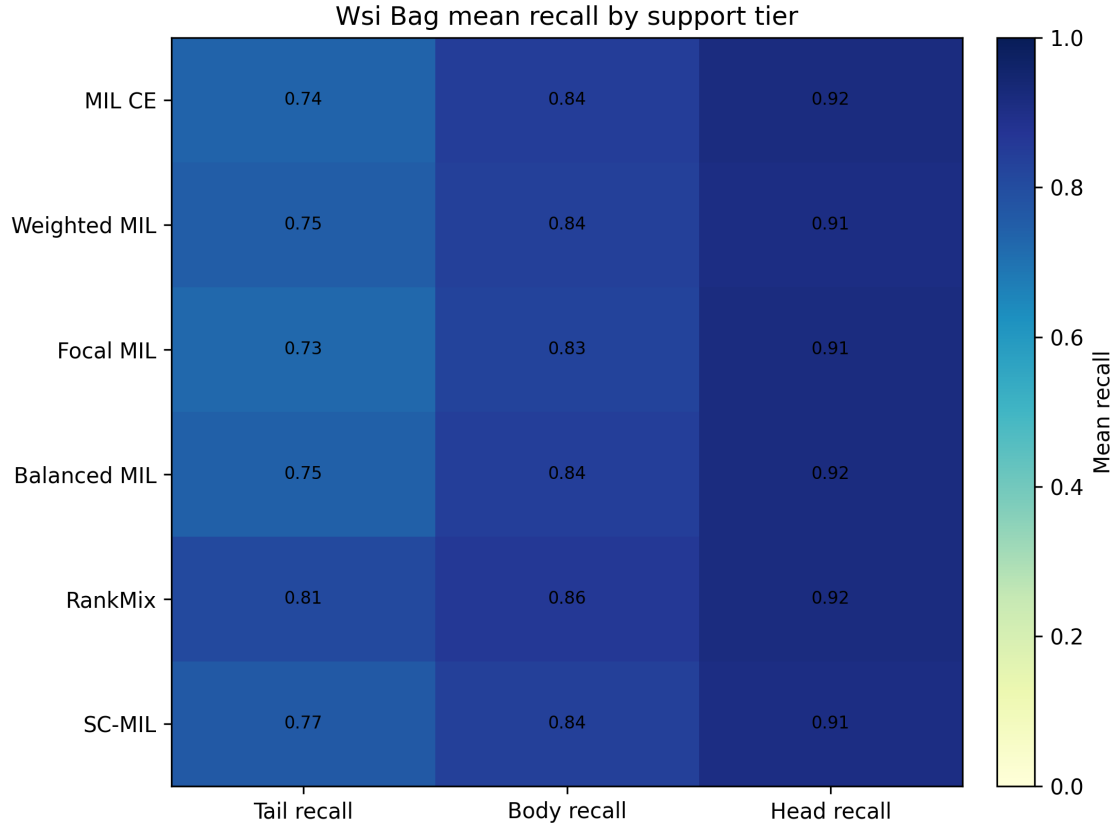


Figure 6: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the long-tailed setting.

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